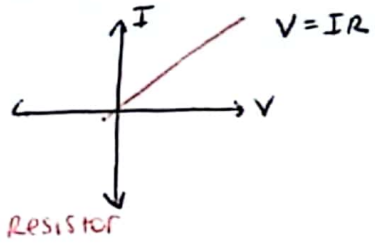
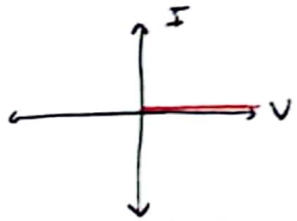
* Electronics 1 *

Passive electronic components: resistors, capacitors, inductors

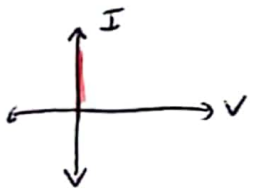
Active electronic components: diodes, transistors



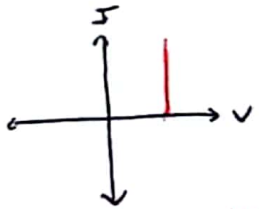
Resistor



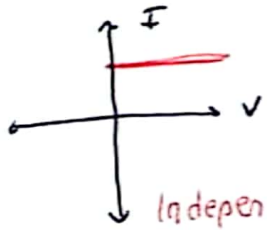
capacitor



Inductor



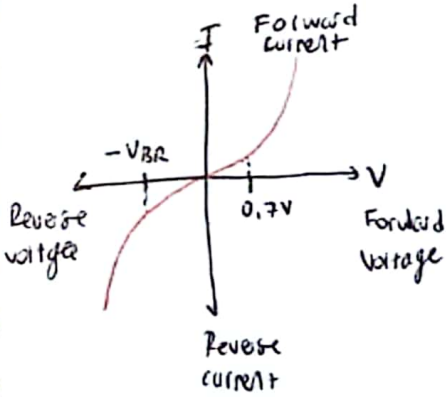
Independent voltage source



Independent current source

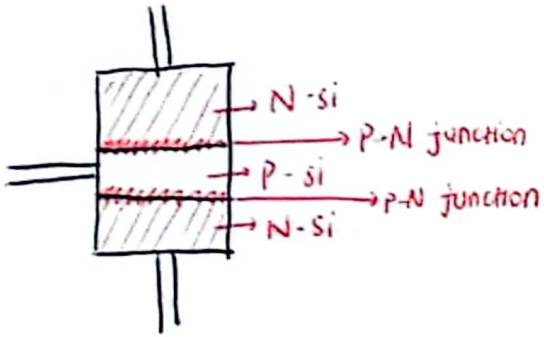
* V-I graphs for passive elements are linear.

* Active element called active because they differentiate from passive element in a way that they can decide how to behave under diff. circumstanties.

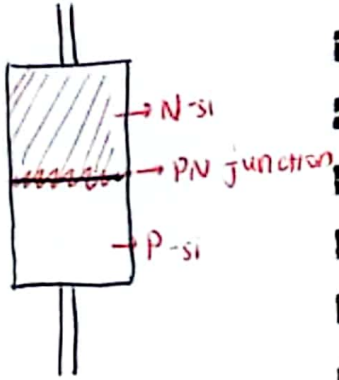


CamScanner ile tarandı

Diode and Transistor Structure

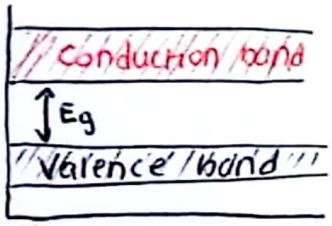


Transistor
Two PN junctions



Diode
(Half Transistor)
One PN junction

Energy Bands



Insulators



Semiconductors



Conductors

E_g ⇒ Band gap energy

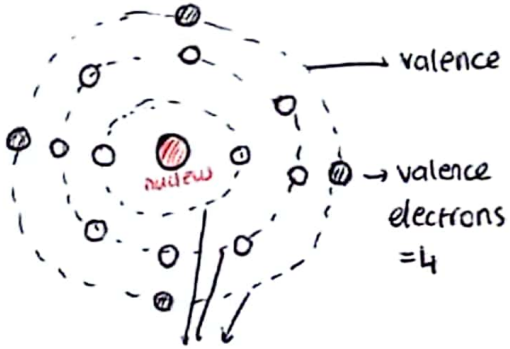
* Diamond $E_g = 9\ eV$
Silicon $E_g = 1.12\ eV$
Copper $E_g = 0\ eV$

Valence band: at T = 0K (atom is not excited) the highest energy band occupied by an electron is called the valence band.

outer shell / valence electrons: electrons that occupied valence band.

conduction band: When atom is excited (absorbs energy) valence electros jump to the higher energy band which is called conduction band.

For Silicon atom ⇒ has 14 electron



energy bands

* We use semiconductors to create active elements. Because active elements are about controlling and you can not get this control mechanism with insulators and conductors.

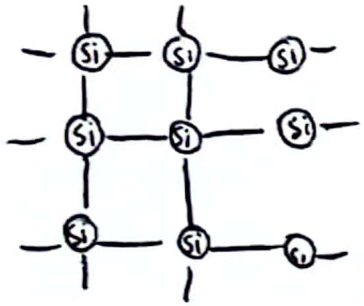
* The mechanism that diode and transistor work is due to the PN junctions.

* We do not use pure silicon because even when you excite the atom free electrons are about $n_i = 1.10^{10}\ cm^{-3}$

↳ $n_i = N_c N_v e^{-(E_c - E_v)/KT} = N_c N_v e^{-E_g/KT}$

density of states

CamScanner ile tarandı

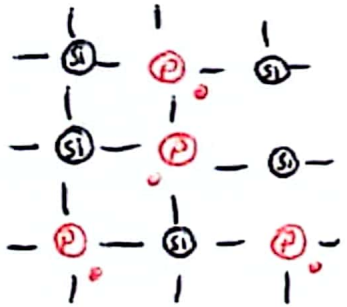


pure silicon

$N_i = 10^{10}$

✖ a good conductors

$N_i = 10^{23}$

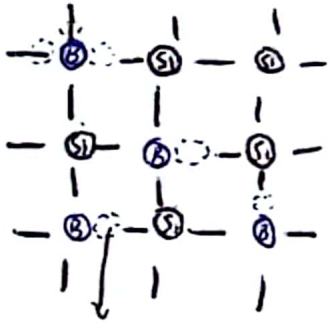


N-type silicon

(more free e^- negative type)

$N_D = 10^{15} \cdot 10^{18}$

donor concentration



hole

P-type silicon

(less free e^- positive type)

$N_A = 10^{13} \cdot 10^{16}$

acceptor concentration

CamScanner ile tarandı

for finding an average of any function.

$$y_{avrg.} = \frac{1}{T}\int_0^{T} y(t)\,dt = \frac{1}{2\pi}\int_0^{2\pi} y(\omega t)\,d(\omega t)$$

* $$i_{avrg.} = \frac{1}{2\pi}\int_0^{2\pi} \frac{24\sin(\omega t) - 12}{100}\,d(\omega t)$$

$$= \frac{1}{2\pi}\int_{Q_1}^{Q_2} \frac{24\sin(\omega t) - 12}{100}\,d(\omega t)$$

to fin Q_1 and Q_2 you

$$V_s(t) = 12 = 24\sin(Q_1)$$

$$Q_1 = \pi/6$$

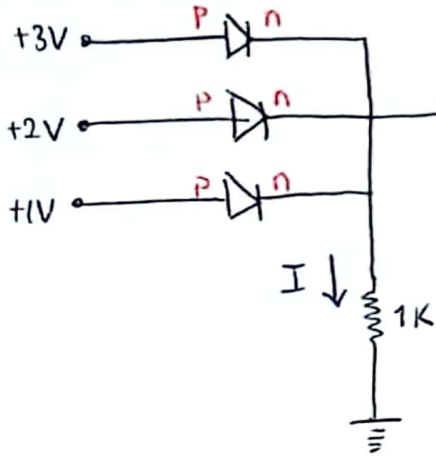
$$Q_2 = \pi - \pi/6 = 5\pi/6$$

$$i_{avrg.} = \frac{1}{2\pi}\int_{\pi/6}^{5\pi/6} \frac{24\sin(\omega t) - 12}{100}\,d(\omega t)$$

from average current you can find average power delivered to the battery.

$$P_{avrg.} = i_{avg} \overbrace{V}^{12V}$$

Example (7)

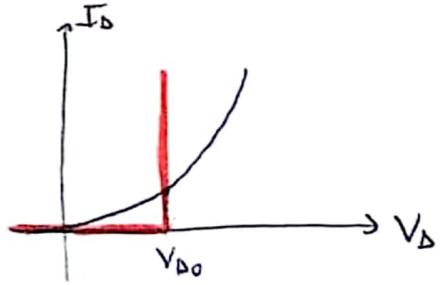


	D_1	D_2	D_3	
$3V = 2V + 1V$	on	on	on	×
$V_o = 0$	off	off	off	×
$V_o = 3$	on	off	off	✓

$V_o = 3V$

$I = \frac{3-0}{1K}$

The Constant-Value Model



for silicon $V_D = 0.7\,V$

* in ideal model when diode on you assume diode as short circuit.

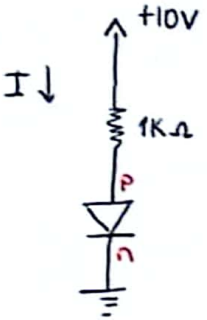
* In constant-value model when diode is on you will assume diode is a constant valued V_{D0} voltage source.

* In both ⇒ diode is off ⇒ open circuit

CamScanner ile tarandı

* $V_p - V_n \geq V_{D0}$

Example (1)



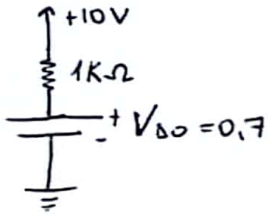
use the constant drop model

$V_{D0} = 0.7V$

$V_p - V_n \geq 0.7V$ open

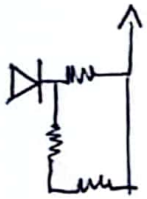
$V_n = 0$

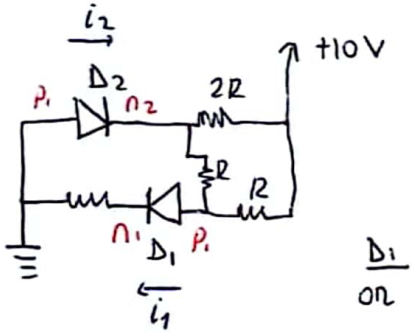
✓ $V_p \geq 0,7V$ diode is on



$$I = \frac{10 - 0.7V}{1K\Omega} = 9.3mA$$

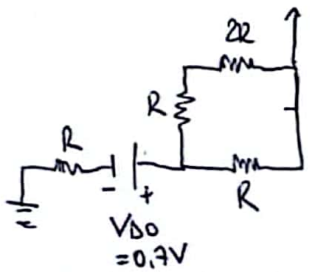
Example (2)





Find current in each diode.

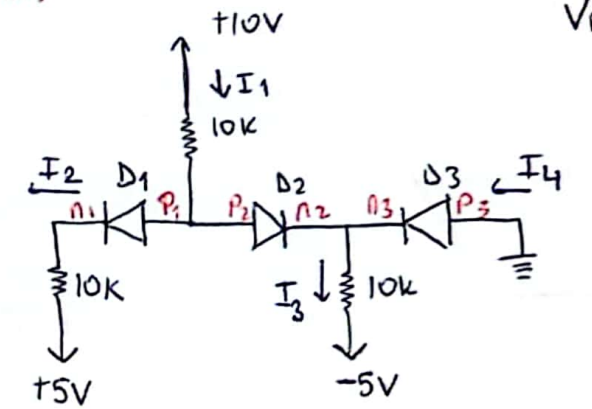
D_1	D_2
on	off



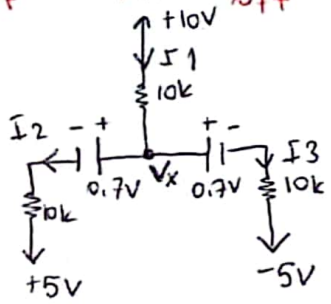
$$\frac{10 - V_{D0}}{9/4R} = i$$

Example (3)

$V_{D0} = 0,7$



D_1	D_2	D_3	assumption
on	on	~~off~~	
off	on	off	



$$\frac{10 - V_x}{10k} = I_1$$

$$\frac{V_x - 0,7 + 5}{10k} = I_3$$

$$\frac{V_x - 0,7 - 5}{10k} = I_2$$

$I_1 = I_2 + I_3$

$$\frac{2V_x - 1.4}{10k} = \frac{10 - V_x}{10k}$$

$3V_x = 11.4$

$V_x = 3.8$

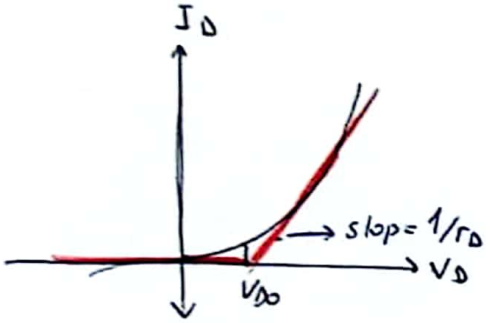
$I_1 = \frac{10 - 3.8}{10k} = 0.62mA$

$I_2 = \frac{3.8 - 5.7}{10} = -0.19mA$) your assumption is wrong

$I_3 = 0.81mA$

CamScanner ile tarandı

The Constant - drop - Resistor model



* $V_p - V_n \geq V_{D0}$

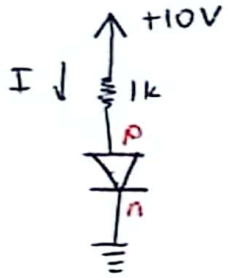
Diode is on

↳ V_{D0} + r_D − (representation)

* $V_p - V_n < V_{D0}$

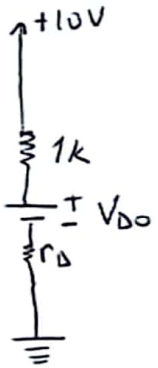
↳ Diode is off

Example



$V_{D0} = 0.7V$ $r_D = 10\Omega$

assume diode is on

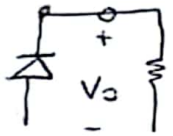


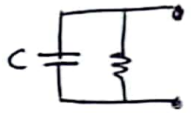
$$I = \frac{10 - 0.7}{1k + 10} = 9.2mA$$

Applications of Diodes

AC to DC converter (DC power supply)

110V AC in → safety features (fuse) → step down voltage (transformer) → 20V - 15V → rectification → smoothing → regulator → DE

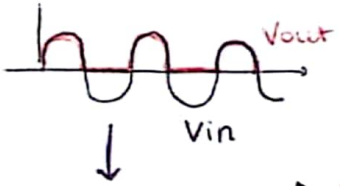


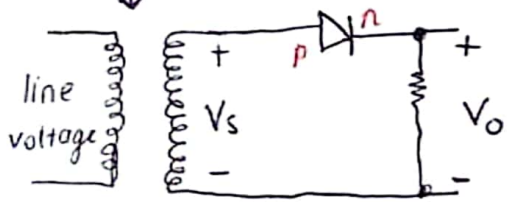




Rectifiers

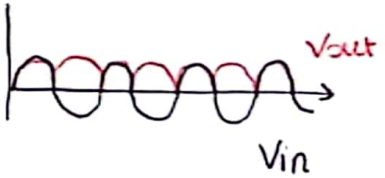
Half - wave rectifier





$V_s(t) = V_{max} \sin \omega t$

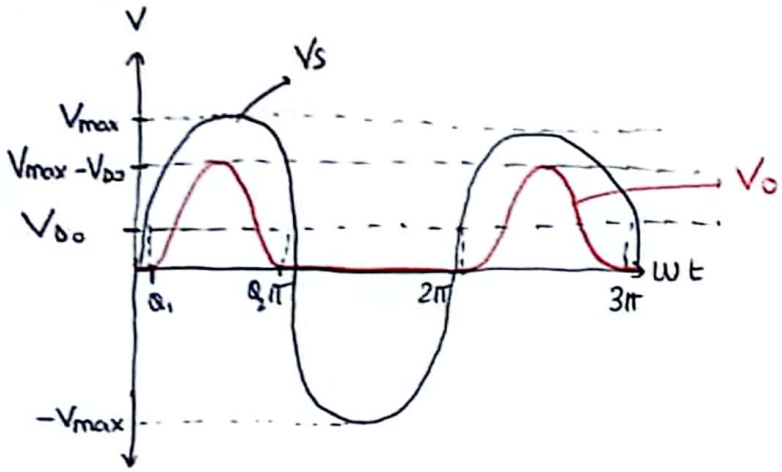
Full - wave rectifier



using constant drop model

$V_{D0} =$

CamScanner ile tarandı



$V_p - V_n \geq V_{DO}$ diode is open

$V_o = V_s(wt) - V_{DO} = V_{max} \sin wt - V_{DO}$

* $V_{o(max)} = V_{max} - V_{DO}$, $i(t) = \frac{V_s - V_{DO}}{R}$

$i_{max} = \frac{V_{max} - V_{DO}}{R}$, $= \frac{V_{max} \sin wt - V_{DO}}{R}$

* $V_{o(avrg)} = \frac{1}{T} \int_0^T V_o(wt)\, dwt) = \frac{1}{T} \int_0^{2\pi} V_o(wt)\, d(wt)$

$= \frac{1}{T} \int_{Q_1}^{Q_2} (V_{max} \sin(wt) - V_{DO})\, d(wt) = \frac{-1}{2\pi} \left[V_{max}(\cos Q_2 - \cos Q_1) + V_{DO}(Q_2 - Q_1) \right]$

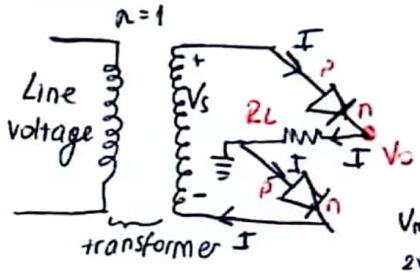
$i_{avrg} = \frac{V_{o\,avrg}}{R}$

$V_s(Q_1) = V_{DO} = V_{max} \sin Q_1$

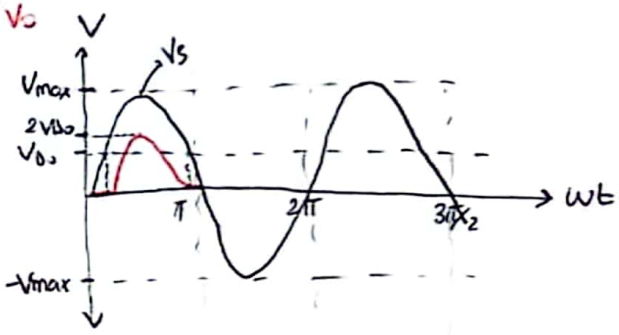
$Q_2 = \pi - Q_1$

V_{max}

Fullwave rectifier (The Bridge Rectifier)

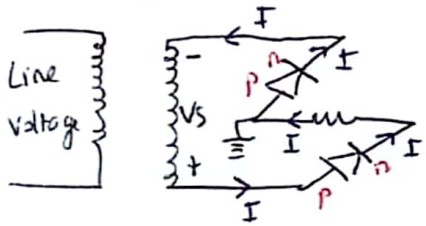


when Vs is "+" this two diodes will be on.

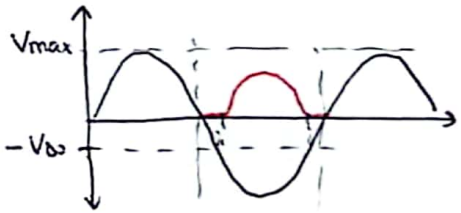


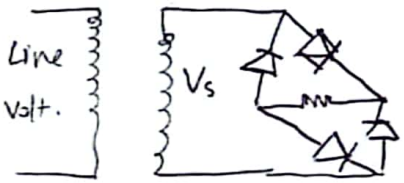
$V_s(t) = V_{max} \sin(wt)$

Since there is ground between two diodes as long as Vs is greater than V_{DO} two diodes will be open.



when Vs is "-" this two diodes will be on.







CamScanner ile tarandı

$$V_{o\,averg} = \frac{1}{2\pi}\int_{0}^{2\pi}[V_s(\omega t) - 2V_{DO}]\,d(\omega t)$$

$$V_{o\,averg} = \frac{2}{2\pi}\int_{\theta_1}^{\theta_2}[V_{max}\sin(\omega t) - 2V_{DO}]\,d(\omega t)$$

assume ideal model ⇒ $V_{o\,ave} = \frac{2V_{max}}{\pi}$

Example

A bridge rectifier is fed by line voltage of 120V (rms) at 60Hz. Transformer has $n = 10:1$ ratio. The load is 1kΩ resistor. $V_{o\,ave}$ = ?

Before transformer ⇒ $V_{max} = \sqrt{2}\,V_{rms} = 120\sqrt{2}$

After transformer ⇒ $V_{max} = \frac{120\sqrt{2}}{10}$

Use ideal model for all diodes

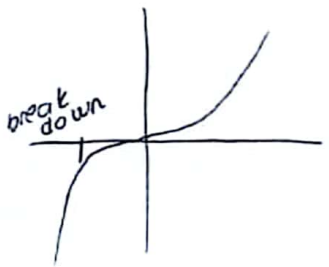
↳ $V_{o\,aver} = \frac{2\cdot(12\sqrt{2})}{\pi} = \frac{24\sqrt{2}}{\pi} = 10.8\,V$

$I_{o\,aver} = \frac{V_{out\,aver}}{R_L} = \frac{10.8\,V}{1k\Omega} = 10.8mA$

$i_{max} = \frac{V_{max}}{R_L} = 17mA$ > in ideal model V_{DO} is 0 so $V_{omax} = V_{max}$

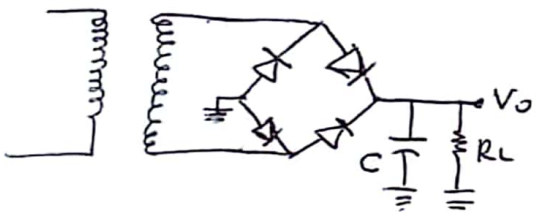
* You should know imax to not burn diodes.

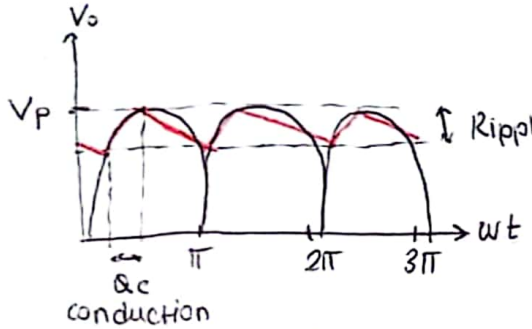
* PIV ⇒ peak inverse voltage ⇒ diode is off



to prevent diodes to break down you should take considir PIV.

Rectifiers with peak detector





C value↑ bigger capacitor

R_L ⇒ load

* You can change C

↳ If you make it so big V_r gets smaller we want that.

V_p = peak value

$V_p = V_{max} - 2V_{DO}$

θ_c being small as possible ⇒ capacitor discharges slow (using less power ~~source~~ from source)

* $V_r = \frac{V_p}{2fRC}$ (ideal model)

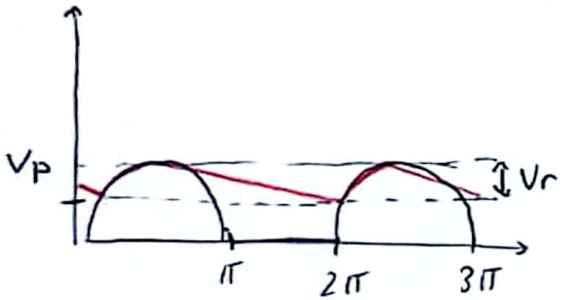
* for $V_{areg.} = V_p - \frac{V_r}{2}$ > approximation because we can not integrate red part.

(ideal model)

CamScanner ile tarandı

* $Q_c = \sqrt{2V_r/V_p}$ * $i_{L(avrg)} = \frac{V_{o\ avrg}}{R_L}$

Half-wave rectifier



* $V_r = \frac{V_p}{fRC}$

* Nobody uses this because capacitor almost fully discharge
↳ not smoth ourput.

Example

Design a peak detector for the bridge rectifier such that ripple voltage is not more than 1V (p-p)

Diodes $V_{D0} = 0.8V$, $R_L = 1k\Omega$

$V_{line} = 120\sin(2\pi 60t)$ $n = 10$

$V_r = \frac{V_p}{2\pi fRC}$

$2\pi 60t = \omega t$ $\omega = 2\pi f$

$V_p = V_{max} - 2V_{D0}$

$f = 60Hz$

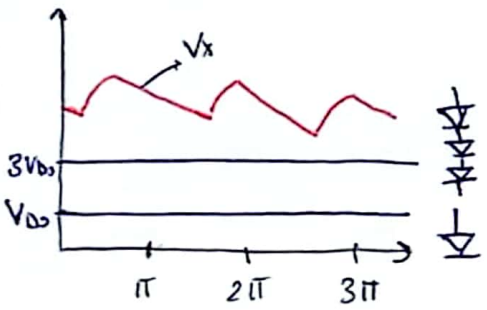
$R = 1k\Omega$

$V_r = \frac{V_{max} - 2V_{D0}}{2\pi f R C}$

$V_{max} = \frac{120}{10} = 12V$

$C = \frac{12 - 2(0.8)}{2(60)(1k)(1)}$ $C = 86.6\mu F$

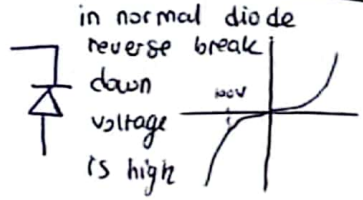
Regulator



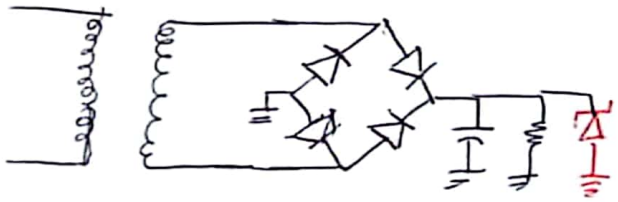
DC power supply

1-) Rectify
2-) Take the peaks only (ripples) Envelop detecto
3-) Regulator

in normal diode reverse break down voltage is high



↳ so your V_x should be high

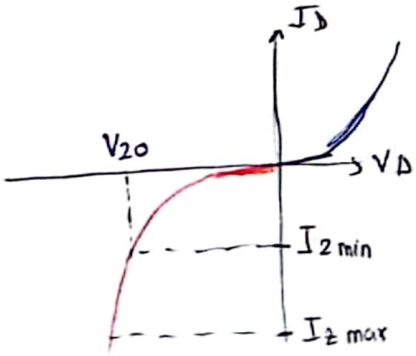


* So we desingn a new diode that its reverse break down voltage is low. ~zener diode~

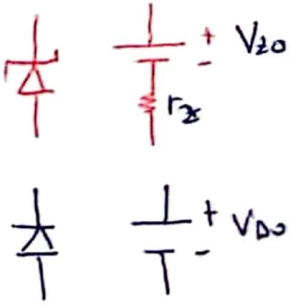
DIN 750 ⇒ 3.6V IN4732 ⇒ 4.7V



* In forward voltage zener diode is just normal diode.



On



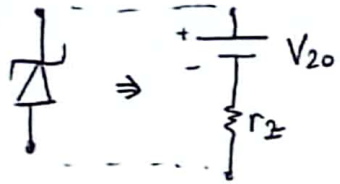
CamScanner ile tarandı

We use zener diode in regulation. So we do care about internal resistor. (slope of the graph backward)

→

* We use I_{zmin} because before that voltage in not exactly constant. Aim to regulate is get constant voltage.

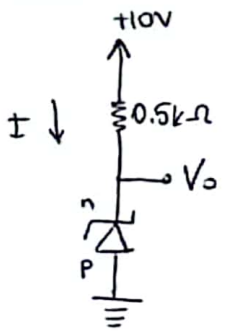
Zener diode Circuit Model in Break down



* $V_z = V_{z0} + I_z . r_z$

Data sheet

Example



Zener $r_z = 20\Omega$

$V_z = 6.8V$ @ $5mA$

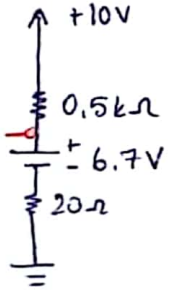
$I_{zk} = 0.2mA$

Find V_o and I

$V_{z0} = V_z - I_z r_z$

$V_{z0} = 6.7V$

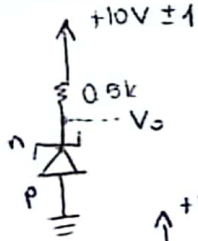
since $V_p - V_n \simeq 10V \Rightarrow$ break down



$I = \frac{10 - 6.7}{20 + 0.5k} = 0.35mA$

$\frac{10 - V_o}{0.5k} = 6.35mA \qquad V_o = 6.83V$

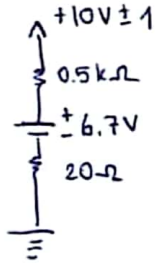
Example



same diode values as previous.

$V_{z0} = 6.7V$ since 11V or 9V is still ≥ 6.7

zener is in break down region



$I = \frac{(+10 \pm 1) - 6.7}{520\Omega}$

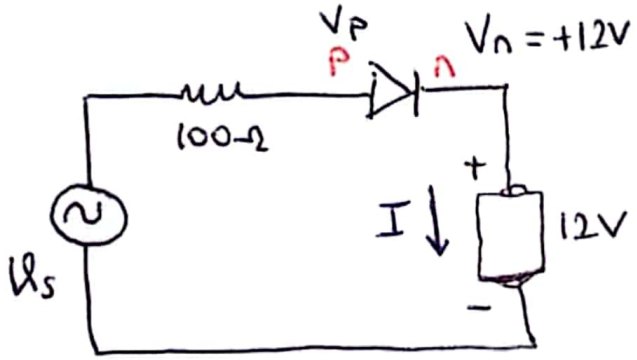
CamScanner ile tarandı

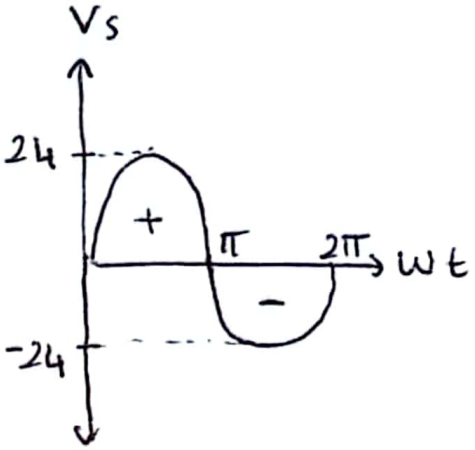
Analysis of Diode circuits

* The Ideal Model

Example (5)

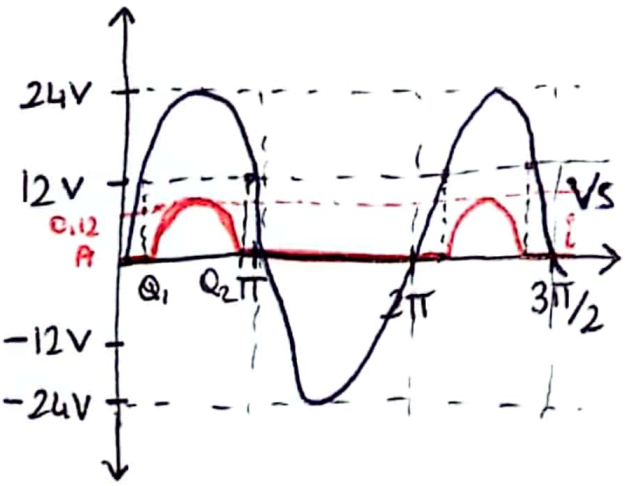
Analyze the circuit below.





$V_s(t) = 24 \sin \omega t$ V

We want to charge the battery. (current should be donward to battery to charge)



$V_p = V_s$

$V_p - V_n \geq 0$ diode on

when diode is on

$$i = \frac{V_s - 12}{R}$$

$V_p - 12 \geq 0$ for diode to be open so when Vs is smaller than 12 diode is off.

$$I_{max} = \frac{24 - 12}{100} = 0.12A$$

CamScanner ile tarandı